

Activity 6 Pre-Lab — Dehydration of Cyclohexanol

[Your name] & [partner]
[MM/DD/YY]

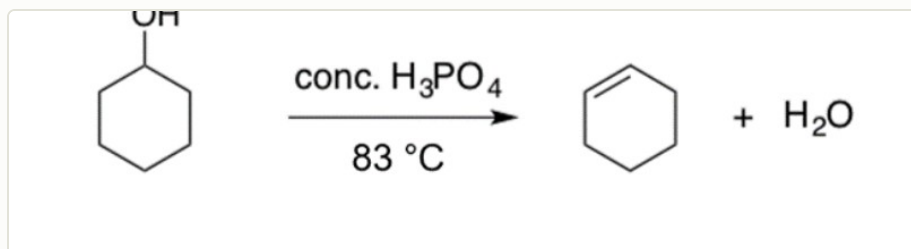
Organic Lab · Synthesis of cyclohexene + purification by simple distillation

■ **Black** = write this in your notebook ■ **Blue** = context, don't copy it Copy the header onto every page. Draw anything in a dashed box yourself.

Purpose

Synthesize **cyclohexene** by the acid-catalyzed **dehydration (E1)** of **cyclohexanol** using 85% phosphoric acid, isolate it by **simple distillation**, then check its purity/identity by TLC, the bromine test, and IR.

🔥 **DRAW** — reproduce this reaction in your notebook (from the procedure)



mechanism = **E1**: protonate the -OH → lose H₂O (carbocation) → lose a β-H → cyclohexene. Also write the conditions (85% H₃PO₄, ≤83 °C, distill).

Chemical Reagents & Safety Table

CONTEXT ▶ Required: cyclohexanol, phosphoric acid, cyclohexene, sodium sulfate, sodium chloride. Typical SDS values — cross-check the SDS tile on Labflow.

Chemical	Role in lab	MW	Amount used	Physical properties	GHS pictograms
Cyclohexanol	Starting material (limiting)	100.16	10 mL (~9.6 g, ~0.096 mol)	Colorless liquid/waxy solid; bp 161 °C; d 0.96 g/mL; mp ~25 °C	 irritant
Phosphoric acid (85%)	Acid catalyst / dehydrating agent	98.00	12 mL	Colorless viscous liquid; d 1.69 g/mL	 corrosive
Cyclohexene	Product	82.14	product (theor. ~0.096 mol)	Colorless liquid; bp 83 °C; d 0.81 g/mL; flash pt -12 °C	    flammable health irritant environ.
Sodium sulfate (anhydrous)	Drying agent	142.04	pea-sized	White solid; hygroscopic	— low hazard
Sodium chloride (sat. aq.)	Brine wash (removes water)	58.44	2 × 10 mL	Colorless aqueous solution	— low hazard

FYI ▶ also used: 10% Na₂CO₃(aq) to neutralize phosphoric acid that distills over (fizzes → CO₂, vent often!); charcoal boiling chips (regular stones dissolve in H₃PO₄).

GRAB FIRST: 100-mL round-bottom (reactor) · 50-mL round-bottom (receiver) · simple-distillation head + condenser + thermometer adapter + thermometer · heating mantle + Variac · lab jack + clamps · ice-water bath · small separatory funnel · 25-mL Erlenmeyer · charcoal boiling chips.

Procedure Plan (left = your plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Synthesis + distillation

1. Measure ~10 mL cyclohexanol into the 100-mL RBF; record exact volume.
2. Add 12 mL 85% H_3PO_4 + a few charcoal boiling chips, swirl to mix. H_3PO_4 is corrosive — gloves; follow instructor's handling.
3. Clamp the reactor to the bars; build a simple-distillation apparatus (still head → thermometer → condenser → 50-mL receiver in an ice-water bath). Heating mantle on a raised lab jack below. Jack must drop enough to slide the hot mantle out.
4. Instructor checks setup → heat with the Variac.
5. Keep the vapor temp $\leq 83^\circ\text{C}$ (adjust Variac / lower mantle). Record the steady distilling temp. 83°C = cyclohexene bp; leaves cyclohexanol + H_3PO_4 behind.
6. Stop (lower jack, mantle off) when 5–10 mL remain in the reactor. **Never distill to dryness.**
7. Neutralize: slowly add 10% $\text{Na}_2\text{CO}_3(\text{aq})$ to the receiver in 5-mL portions, swirl till fizzing stops → transfer to sep funnel. Vent often in the hood — CO_2 builds pressure.
8. Wash the organic layer 2 × 10 mL saturated NaCl; drain off the aqueous each time. Product = organic (top) layer.
9. Transfer product to a 25-mL Erlenmeyer; dry over a pea-sized amount of anhydrous Na_2SO_4 until no longer cloudy.
10. Decant the clear cyclohexene into a pre-weighed flask; weigh → mass of product → % yield.

Tests (do all before discarding product)

11. TLC (30% EtOAc/hexanes) · Bromine test (Br_2 decolorizes → alkene) · IR (look for =C-H ~3050, C=C ~1650; loss of broad O-H).

WRITE

$$\% \text{ yield} = (\text{actual mol cyclohexene} \div \text{theoretical mol}) \times 100$$

Experiment Notes (in lab)

Exact cyclohexanol volume: _____ mL

Steady vapor temp: _____ $^\circ\text{C}$

Distillation observations:

Workup — layers / venting / bubbling:

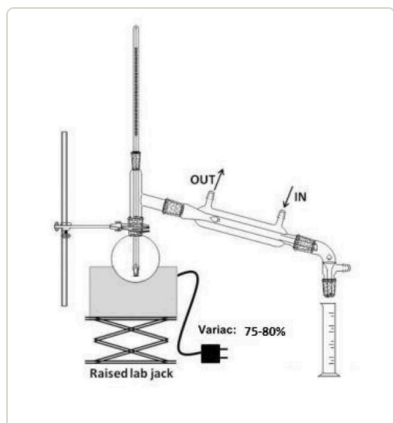
Product appearance (clear? turbid?):

Mass flask: _____ g flask+product: _____ g

Mass product: _____ g % yield: _____

Bromine test / TLC R_f / IR peaks:

Setup — draw this



simple distillation apparatus

CONTEXT (don't copy) ▶

Why 83°C cap? cyclohexene boils at 83°C ; cyclohexanol (161°C) and H_3PO_4 stay behind, so the distillate is mostly pure product.

Why brine + Na_2SO_4 ? saturated NaCl pulls water out of the organic layer; anhydrous Na_2SO_4 then dries the last traces (cloudy → clear).

Clean-up & Conclusions

- Acidic pot residue → neutralize/dispose per instructor; organic waste for cyclohexene after tests; aqueous washes → aqueous waste; return glassware; do your bench task.
- **Conclusions** (~4 sentences [after lab](#)): your % yield and a reason it wasn't 100% (product left in pot, evaporation, incomplete rxn); what the bromine test + IR C=C confirmed; and how distillation purified the product.

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